Lab 3 Rubric: Projectile Motion

Out of 100 points. This is the rubric on the LB3-M2 assignment in Brightspace. Lab 3 page · Word copy · PDF copy

Use this rubric as your writing guide: the criteria follow the lab report template top to bottom. Full-credit descriptors say exactly what a complete answer contains.

Grammar, spelling, and formatting problems are assessed 1 point per occurrence. Deadline: a report submitted by the deadline may be revised and resubmitted up to two more times. A report that arrives after the deadline but before grades are posted is graded as on time but cannot be revised; one that arrives after grades are posted is late, may take a penalty, and gets no detailed feedback.

The phrase “human error” earns no credit anywhere in the report. A percent difference under 10% may be attributed to ordinary experimental uncertainty, provided you say so explicitly. Over 10%, you must name specific error sources and quantify them.

Criterion	Pts	Complete	Strong	Developing	Needs work	Missing
Heading	3	**3** Names, course and section, lab title and date, all present and correctly formatted.	**2** One item missing or misformatted.	**2** Template placeholders or brackets left in the heading, or the section number missing.	**1** Several items missing.	**0** No heading block.
Abstract	7	**7** Four sentences or fewer. States what was measured, gives v_0 with its uncertainty and the percent differences, and says whether the prediction held. Nothing else.	**6** All content present but runs long or includes method detail.	**4** Reports v_0 or the percent differences but not both, or gives v_0 with no uncertainty.	**2** Describes the lab but reports no numbers.	**0** No abstract.
Introduction	10	**10** Derives $d = v_0\sqrt{(2h/g)}$ line by line from the two axis equations, eliminating the shared time; every variable defined; states a hypothesis that can later be confirmed or rejected.	**8** Derivation present but a step or a variable definition is missing.	**6** The derivation is sketched rather than worked line by line, or a variable is left undefined, though the hypothesis is present.	**3** The equation is quoted rather than derived, or there is no hypothesis.	**0** No introduction.
Experimental: Materials & Setup	8	**8** Apparatus described in prose; a photo of the actual setup AND a labelled sketch, with both measurement reference points marked — where h is measured from and to, and where d is measured from and to.	**7** Photo and sketch present, but a reference point is unmarked or a component unlabelled.	**5** Apparatus described in prose with a photo or a sketch, but not both, or a measurement reference point is left unmarked.	**3** Bare equipment list, or only one of photo and sketch.	**0** No materials or setup description.
Experimental: Procedure	8	**8** Written in own words as prose, reproducible, and states how the launcher was levelled and how the plumb point was found.	**7** Reproducible and in the student's own words, but one detail is thin.	**5** Reproducible, but missing the levelling or the plumb-point detail.	**3** Copied or closely paraphrased from the handout.	**0** No procedure.
Results: Data Tables	12	**12** Calibration table of five or more trials with mean and SEM; table of every launch height with predicted distance, mean measured distance, SEM and percent difference. All numbered, captioned, units and consistent significant digits, referenced in the text.	**10** Both tables present; minor issues with captions, units or significant digits.	**7** One of the two tables is complete but the other is missing or thin, or the calibration run has fewer than five trials.	**4** A table missing, or fewer than five trials at the calibration height.	**0** No data tables.

Criterion	Pts	Complete	Strong	Developing	Needs work	Missing
Results: Graphs	**9**	**9** One graph carrying both: the theory as a smooth dashed curve CALCULATED from v_0, and the measured means as discrete markers with vertical error bars. Numbered as a Figure, captioned below, described in the text.	**7** Both series present and correctly calculated; a caption or the in-text description is incomplete.	**5** Both series present, but the theory curve is fitted to the data rather than calculated from v_0, or the error bars are missing.	**3** Only one series plotted, or the data joined dot-to-dot.	**0** No graph.
Results: Sample Calculations	**13**	**13** One worked sample of each type, with units at every step: fall time, v_0 from h and $\bar{d}$, uncertainty propagation in v_0, predicted d at a new height, mean and SEM, percent difference.	**11** All types present with minor unit or rounding slips.	**8** Four or five of the six types worked; one or two missing.	**4** Some types missing, or units given only in the final answer.	**0** No sample calculations.
Discussion	**15**	**15** Opens with the measured $v_0 \pm$ uncertainty; states whether the hypothesis is confirmed and argues it with the 10% rule; addresses the PATTERN in the percent differences across heights, not only their average; states plainly for each height whether the ball landed in the cup, including the misses.	**12** Most of the above; one element thin or missing.	**9** Opens with a result and interprets it, but two or three of the required moves are absent — the 10% argument, the pattern in the percent differences across heights, or the per-height record of whether the ball landed in the cup.	**5** Restates the results without interpreting them, or reports only an average percent difference.	**0** No discussion.
Additional Questions	**5**	**5** All three Additional Questions (AQ1–AQ3) answered in paragraphs, in a dedicated section.	**4** One of the three answered thinly.	**3** Answered inside another section rather than under their own heading.	**2** Answered as a bare list, or one of the three skipped.	**0** Not answered.
Conclusion	**5**	**5** Three to four sentences: numerical result with percent difference, a definitive verified / not-verified statement, and one concrete improvement to the measurement.	**4** Present but missing the numbers or the verdict.	**3** Present with the numbers but no definitive verified / not-verified verdict, or no improvement named.	**2** Present but vague, or a repeat of the discussion.	**0** No conclusion.
References	**5**	**5** At least one source beyond the textbook; author, title and date given for each.	**4** Multiple references but formatting incomplete.	**3** Textbook and handout only, properly formatted.	**2** Handout only.	**0** No references.
Total	**100**					

Overall: Complete 90+ · Nearly there 75+ · Needs work 50+ · Missing below 50.

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